

Work Toward Your Project — Milestone 5

In this milestone, you will complete the *quality phase* of your system—evaluating performance, auditing fairness, and explaining model behavior—then finalize and present your project. The core scope in the project brief is: compute intrinsic metrics, run at least one fairness probe, generate a local explanation for a correct and an incorrect example, and synthesize findings into your final deliverables.

What you will deliver (end of Week 5)

- **Final report or slide deck (8–10 pages)** synthesizing design choices, metrics, fairness, interpretability, and next-step recommendations.
- **A runnable notebook/script** that reproduces your core experiments.
- **Project presentation recording (8–12 minutes)** showing (a) your system running, (b) results from all milestones, and (c) live outputs for key evaluations (see “Recording specs” below).

Part A — Intrinsic Evaluation (quality as numbers)

Goal: Report the right metrics for your task and interpret them.

1. **Choose task-appropriate metrics.**
 - Classification: Accuracy, Precision/Recall/F1 (+ confusion matrix).
 - Sequence labelling: entity-level P/R/F1.
 - Seq2seq: BLEU/ROUGE (and a short qualitative sample table).
2. **Slice briefly.** Add one simple slice (e.g., short vs long inputs) to reveal error patterns.
3. **Write 3–4 lines:** What the numbers *say* (and *don’t say*) about user impact.

Tie back to your earlier baselines and transformer upgrade to show progress across milestones.

Part B — Fairness Probe (who bears the mistakes?)

Goal: Quantify a concrete harm and report a gap.

1. **Declare the harm** (e.g., over-moderation).
2. **Run one probe** aligned to your task, e.g., **FPR gap by identity slice** (toxicity), or an **association probe** (WEAT/StereoSet) for embeddings/generators.
3. **Report** per-group table + **gap** (max–min) and **worst-group** value.
4. **One-line action threshold** (e.g., “>3 pp gap ⇒ mitigation next release”).
5. **Interpret in 3–4 lines** (why this matters for users or product).

A fairness probe is explicitly required at this stage.

Part C — Single-Prediction Explanations (why the model decided so)

Goal: Produce *local* explanations for one correct and one incorrect case.

1. **Method:** Attention map or LIME/SHAP (choose one consistent method).
2. **Show** top contributing tokens/features and a short narrative (≤ 80 words) per example: what evidence the model used; whether it's spurious or sensible.
3. **Faithfulness quick check (optional):** mask/replace a top token and note Δ in predicted probability.

A local explanation for **both** a correct and an incorrect example is part of Milestone 5.

Part D — Synthesis & Next Steps

Goal: Turn evidence into decisions.

Create a one-page “Findings” section (or slide) with:

- **Primary success metric** and headline numbers.
 - **Fairness headline:** gap + worst-group.
 - **Explanation insight:** one spurious cue you will address.
 - **One concrete mitigation** (e.g., threshold by slice, data augmentation, domain adaptation) and how you will **re-measure** after applying it.
-

Packaging & Reproducibility (checklist)

- **Repo/notebook** runs end-to-end on a small sample without manual tweaking (README with env + run steps).
 - **Data note:** where to fetch or how to sample (if licensing).
 - **Seeds & versions:** fix random seeds; list key library versions.
 - **Outputs folder:** confusion matrix image, small qualitative table (e.g., 5 references vs predictions), explanation screenshots.
 - **Provenance:** link back to earlier milestones (M1–M4) to show progression across the course-long project.
-

Presentation recording — specs & outline

Length: 8–12 minutes. **Format:** screen recording with voiceover; camera optional.

What to show (in order):

1. **Problem & data (≤60s):** task, dataset, users.
2. **System overview (≤60s):** current model vs earlier baselines (one slide).
3. **Demo (2–3 min):** run the notebook/script on 2–3 examples; show a live prediction or pipeline output.
4. **Metrics (1–2 min):** headline intrinsic scores + one slice; what improved since M2/M3.
5. **Fairness (1–2 min):** primary gap, worst-group, and what that implies.
6. **Explainability (1–2 min):** the two local explanations (one correct, one incorrect) and the key takeaway.
7. **Decision & next step (≤60s):** one mitigation and how you will re-measure.

Recording tips: speak to *why each number matters*; keep figures legible; show one live cell execution so reviewers see it run.

Submission format (LMS)

Upload a single ZIP containing:

- `report.pdf` **or** `slides.pdf` (8–10 pages).
- `notebook.ipynb`.
- `recording.mp4` (8–12 min).
- `README.md` with run steps and links to any large data assets.